



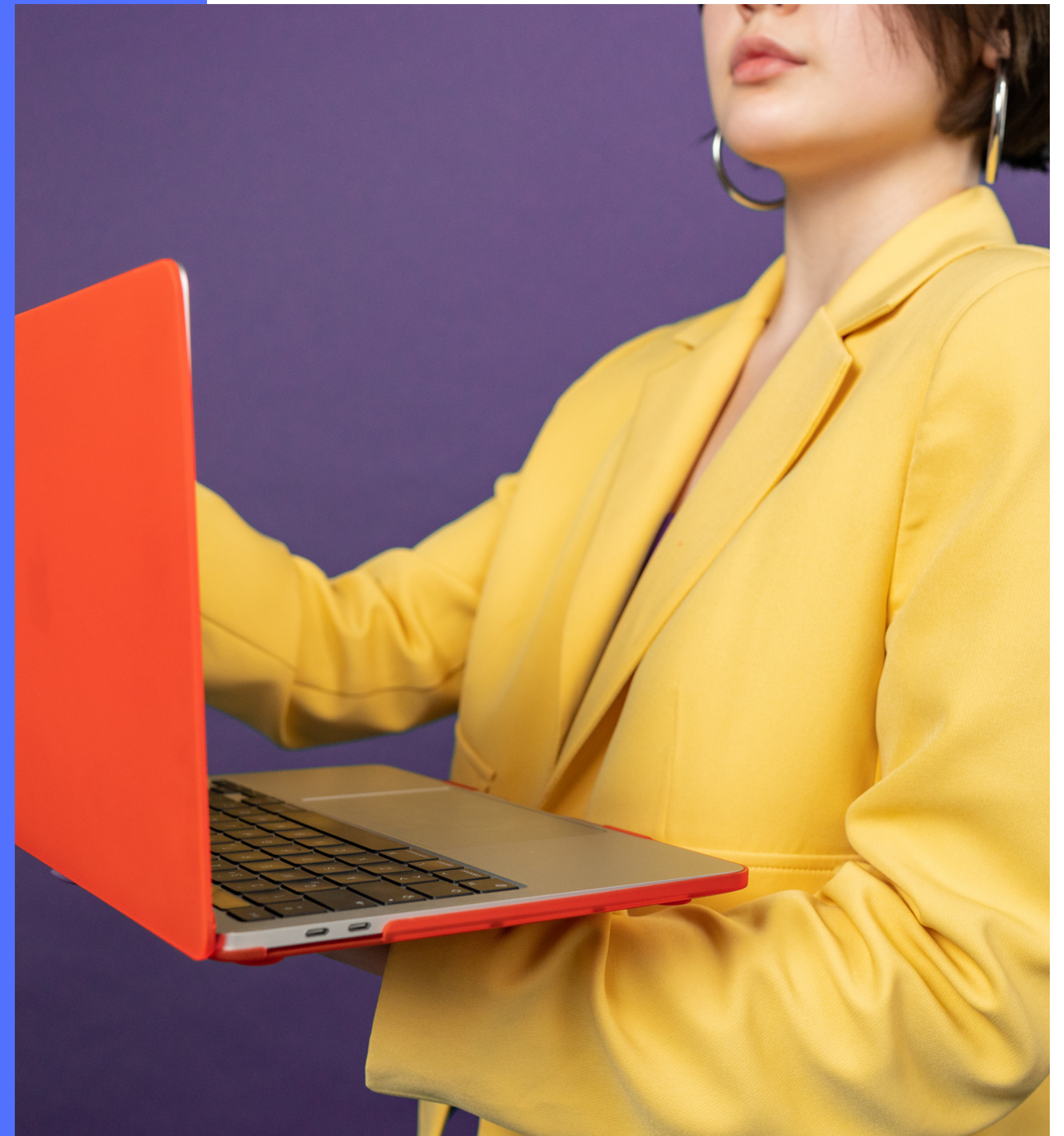
WGU MASTER OF SCIENCE, DATA ANALYTICS

DATA ACQUISITION

PRESENTATION BY
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RESEARCH QUESTION

AMONG FEMALE CUSTOMERS
35 AND UNDER,
HOW MANY USE OUR TECH
SUPPORT SERVICES?



QUESTION DEFINITIONS

FEMALE CUSTOMERS UNDER 35

Female = self-identify as 'Female'

Noting, there is a *non-binary* option

USE OUR TECH SUPPORT SERVICES

Answer 'Yes' under tech_support services

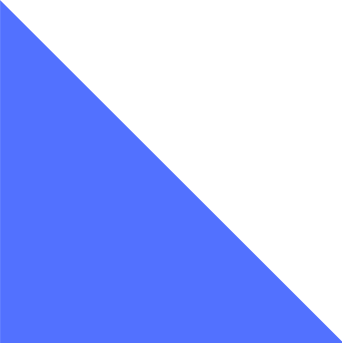


CREATING THE SERVICES TABLE

Switch into pgAdmin...

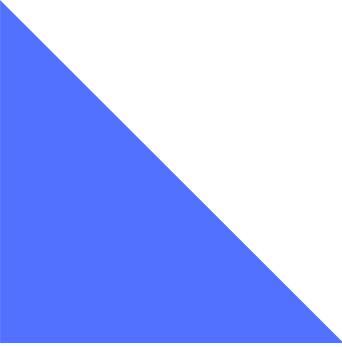
```
CREATE TABLE public.services
(
customer_id text NOT NULL,
internet_service varchar(20) NOT NULL,
phone varchar(3) NOT NULL,
multiple varchar(3) NOT NULL,
online_security varchar(3) NOT NULL,
online_backup varchar(3) NOT NULL,
device_protection varchar(3) NOT NULL,
tech_support varchar(3) NOT NULL,
PRIMARY KEY (customer_id),
CONSTRAINT customer_id_fkey FOREIGN KEY (customer_id)
REFERENCES public.customer (customer_id) MATCH SIMPLE
ON UPDATE NO ACTION
ON DELETE NO ACTION
NOT VALID
)
TABLESPACE pg_default;

ALTER TABLE public.services
```





BREAKING DOWN THE MAIN SQL QUERY



```
SELECT
/* Select customer age */
    c.age AS customer_age,
/* Use the SUM function with a CASE statement to tally how many women under
the age of 35 have tech support */
    SUM(CASE WHEN c.gender = 'Female' AND c.age <= 35 AND s.tech_support =
'Yes' THEN 1 ELSE 0 END) AS female_customers_with_tech_support,
/* Use the COUNT function with a CASE statement to count the total number
of women under 35 with or without tech support */
    COUNT(CASE WHEN c.gender = 'Female' AND c.age <= 35 THEN c.customer_id
END) AS total_female_customers,
/* Use the COUNT and SUM functions together with a CASE statement to
calculate the percentage of female customers under 35 who have tech support
*/
    CASE WHEN COUNT(CASE WHEN c.gender = 'Female' AND c.age <= 35 THEN
c.customer_id END) > 0
        THEN (CAST(SUM(CASE WHEN c.gender = 'Female' AND c.age <= 35 AND
s.tech_support = 'Yes' THEN 1 ELSE 0 END) AS DECIMAL) / COUNT(CASE WHEN
c.gender = 'Female' AND c.age <= 35 THEN c.customer_id END)) * 100
        ELSE 0
    END AS percentage_female_customers_with_tech_support
FROM
    customer AS c
/* Use a Left Join to link the 2 tables, customer and services on
customer_id */
LEFT JOIN
    services AS s ON c.customer_id = s.customer_id
/* Filter customer records to extract those where the age is less than or
equal to 35 */
WHERE
    c.age <= 35
/* Group and Order By customer age in descending order, going from oldest
to youngest customers */
GROUP BY
    c.age
ORDER BY
    c.age DESC;
```

GET CUSTOMER AGE

```
SELECT  
    c.age AS customer_age,
```

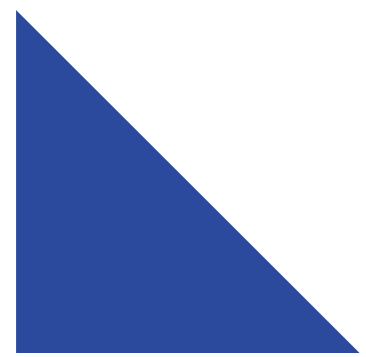
customer_age	
integer	
34	
33	
32	
31	
30	
29	
28	
27	
26	
25	
24	
23	
22	
21	
20	
19	
18	



female_customers_with_tech_support	bigint
	21
	21
	30
	25
	21
	28
	21
	27
	35
	21
	32
	25
	22
	29
	32
	30
	24

GET FEMALE CUSTOMERS WITH TECH SUPPORT

```
SUM(CASE WHEN c.gender = 'Female' AND c.age <= 35 AND s.tech_support = 'Yes' THEN 1 ELSE 0 END) AS female_customers_with_tech_support,
```





GET TOTAL FEMALE CUSTOMERS

```
COUNT(CASE WHEN c.gender = 'Female' AND c.age <= 35 THEN c.customer_id
END) AS total_female_customers,
```



total_female_customers	
bigint	
	67
	60
	77
	71
	66
	83
	72
	69
	75
	66
	80
	68
	60
	71
	72
	81
	65

GET PERCENTAGE OF FEMALE CUSTOMERS WITH TECH SUPPORT

```
CASE WHEN COUNT(CASE WHEN c.gender = 'Female' AND c.age <= 35 THEN  
c.customer_id END) > 0  
    THEN (CAST(SUM(CASE WHEN c.gender = 'Female' AND c.age <= 35 AND  
s.tech_support = 'Yes' THEN 1 ELSE 0 END) AS DECIMAL) / COUNT(CASE WHEN  
c.gender = 'Female' AND c.age <= 35 THEN c.customer_id END)) * 100  
    ELSE 0  
END AS percentage_female_customers_with_tech_support
```

percentage_female_customers_with_tech_support	numeric
31.3432835820895	
35.0000000000000	
38.9610389610389	
35.2112676056338	
31.8181818181818	
33.7349397590361	
29.1666666666666	
39.1304347826086	
46.6666666666666	
31.8181818181818	
40.0000000000000	
36.7647058823529	
36.6666666666666	
40.8450704225352	
44.4444444444444	
37.0370370370370	
36.9230769230769	

**FROM
CUSTOMER TABLE**

```
FROM  
    customer AS c
```

JOIN TABLES ON CUSTOMER_ID

```
LEFT JOIN
```

```
services AS s ON c.customer_id = s.customer_id
```

FILTER RECORDS

```
WHERE  
    c.age <= 35
```

GROUP & ORDER RECORDS

```
GROUP BY
```

```
    c.age
```

```
ORDER BY
```

```
    c.age DESC;
```

RESULTS OF THE QUERY

Data Output		Explain	Messages	Notifications	
customer_age integer		female_customers_with_tech_support bigint		total_female_customers bigint	percentage_female_customers_with_tech_support numeric
1	35	22	79	27.84810126582278481000	
2	34	21	67	31.34328358208955223900	
3	33	21	60	35.00000000000000000000	
4	32	30	77	38.96103896103896103900	
5	31	25	71	35.21126760563380281700	
6	30	21	66	31.81818181818181818200	
7	29	28	83	33.73493975903614457800	
8	28	21	72	29.16666666666666666700	
9	27	27	69	39.13043478260869565200	
10	26	35	75	46.66666666666666666700	
11	25	21	66	31.81818181818181818200	
12	24	32	80	40.00000000000000000000	
13	23	25	68	36.76470588235294117600	
14	22	22	60	36.66666666666666666700	
15	21	29	71	40.84507042253521126800	
16	20	32	72	44.44444444444444444400	

UPDATE RECOMMENDATIONS

TIME PERIOD

Update database every month

GOING FORWARD

Review data, again, after 2 years

-END-